

Design of Components

Smart Student Club Management System

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1. User Authentication Component

Purpose

The User Authentication Component verifies user identity and grants access to the system based on predefined roles. It serves as the entry point for all users and ensures that only authorized users can access system functionalities.

Responsibilities

- Authenticate users using login credentials.
- Identify the user's role.
- Redirect users to the appropriate dashboard.
- Enable additional dashboard components when a student's role changes (e.g., Volunteer).

Inputs

- User ID / Student ID
- Password

Outputs

- Successful or failed authentication.
- User dashboard based on assigned role.

Interacts With

- Dashboard Component

Functional Behaviour

- User enters Student ID/User ID and password.
- System validates the credentials.
- User role is identified.
- Appropriate dashboard is loaded.
- Additional dashboard modules are enabled based on approved roles (e.g., Volunteer).
- Invalid login attempts display an authentication error.

2. Dashboard Component

Purpose

The Dashboard Component provides users with a personalized interface by displaying features and information according to their assigned roles.

Responsibilities

- Display available modules according to user role.
- Enable additional components after approvals.
- Provide quick access to club activities, events, tasks, reports, and certificates.

Functional Behaviour

Student Dashboard

- Browse clubs.
- Browse events.
- Register for events.
- Apply for club membership.
- Apply as a volunteer.
- View certificates.

Volunteer Dashboard

- Pending Tasks.
- Ongoing Tasks.
- Completed Tasks.
- Upload Proof of Work.
- View volunteer certificates.

Club Head Dashboard

- Club management.
- Event proposal preparation.
- Membership request management.
- Volunteer recruitment.
- Approve/reject volunteer applications.
- Create logical event domains (Operations, Logistics, Marketing, Design, Sponsorship, etc.).
- Create and assign tasks to volunteers under different domains.
- Assign Field Monitors to supervise physical work.
- Track task progress using Kanban Board.
- Verify uploaded proof of work.
- View event progress dashboard.

- Send messages or reminders to volunteers and field monitors.
- Request certificate generation after event completion.
- View attendance statistics.
- Event progress.

Club President Dashboard

- Club proposal review.
- Event approval.
- Performance monitoring.

Faculty Coordinator Dashboard

- Club approval.
- Event approval.
- Venue and asset approval.
- Reports and analytics.

3. Club Management Component

Purpose

Manages the creation of clubs and membership within clubs.

Responsibilities

- Process new club creation requests.
- Process club membership requests.
- Activate "My Clubs" after membership approval.

Functional Behaviour

Club Creation

- Club President submits a new club proposal.
- Proposal is forwarded to the Faculty Coordinator.
- Faculty Coordinator approves or rejects the proposal.
- Approved clubs become available in the system.

Membership Management

- Student submits a club membership request.
- Club Head reviews the application.
- Application is approved or rejected.
- Approved club appears in the student's **My Clubs** section.
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4. Event Management Component

Purpose

Manages the complete event lifecycle from proposal preparation to publication.

Responsibilities

- Event proposal preparation.
- Event approval process.
- Event publication.
- Event registration.

Functional Behaviour

Event Proposal

- Club Head creates a new event proposal.
- Club Head enters event details, schedule, venue requirements, asset requirements, volunteer requirements, and budget estimation.
- Proposal is submitted to the Club President.
- Club President reviews the proposal.
- Approved proposal is forwarded to the Faculty Coordinator.
- Faculty Coordinator approves or rejects the proposal.
- Approved events are published.
- Venue and asset reservation process is initiated.

Event Registration

- Student browses available events.
- Student views event details.
- System verifies registration availability.
- Student registers for the event.
- QR code is generated.
- Registered event appears on the student's dashboard.

5. Venue and Asset Management Component

Purpose

Manages venue allocation and institutional assets required for events.

Responsibilities

- Display resource availability.
- Allocate venues.
- Allocate assets.
- Reserve approved resources.
- Release rejected allocations.
- Manage asset lifecycle.

Functional Behaviour

- Club Head selects required venues and assets during proposal preparation.
- System checks resource availability.
- Available resources are temporarily reserved.
- Faculty Coordinator approves the event.
- Resources become officially reserved.
- If another event requests the same resource at the same time, conflict detection is performed.
- Rejected proposals release reserved resources.
- Resource status is updated throughout the event lifecycle.

6. Volunteer Management Component

Purpose

Manages volunteer recruitment for approved events.

Responsibilities

- Publish volunteer recruitment.
- Receive volunteer applications.
- Club Head reviews applications.
- Approve or reject volunteers.
- Assign volunteers to event domains.
- Assign Field Monitors.
- Enable Volunteer Dashboard.

Functional Behaviour

- Event is approved.
- Volunteer recruitment is published.
- Students apply as volunteers.
- Club Head reviews volunteer applications.
- Applications are approved or rejected.
- Approved volunteers receive Volunteer Dashboard access.
- Club Head assigns volunteers to event domains.
- Club Head appoints Field Monitors for physical supervision.

7. Task Management Component

Purpose

Supports task assignment and monitoring during event execution.

Responsibilities

- Create event domains.
- Create tasks.
- Assign volunteers.
- Assign Field Monitors.
- Track task progress.
- Maintain Kanban board.
- Send reminders.
- Reassign tasks.
- Monitor work completion.

Task States

- Pending
- Accepted
- In Progress
- Under Review
- Completed
- Revision Required

Functional Behaviour

- Club Head creates logical work domains.
- Club Head creates tasks under each domain.
- Volunteers are assigned to tasks.
- Field Monitors are assigned where required.
- Volunteers receive task notifications.
- Volunteers accept assigned tasks.
- Accepted tasks move to **In Progress**.
- Field Monitors monitor work on-site.
- Club Head tracks task progress through the Kanban Board.
- Volunteers upload proof after task completion.
- Club Head reviews submissions.
- Tasks are marked as Completed or returned for revision.
- Club Head may send reminders or reassign tasks when necessary.

8. Proof of Work Verification Component

Purpose

Verifies submitted work before marking tasks as complete.

Responsibilities

- Receive uploaded proof.
- Review submissions.
- Request revisions.
- Approve completed work.

Functional Behaviour

- Volunteer uploads proof of completed work.
- Club Head reviews the submitted proof.
- System verifies task completion status.
- Club Head approves the submission or requests revisions.
- Approved tasks are marked as Completed.
- Revision requests return the task to the volunteer.

9. QR Attendance Component

Purpose

Records participant attendance electronically.

Responsibilities

- Verify participant registration.
- Record attendance.
- Maintain live attendance count.

Functional Behaviour

- Participant arrives at the event.
- Volunteer scans the participant's QR code.
- System verifies registration.
- Attendance is recorded.
- Live attendance count is updated.
- Club Head views attendance statistics through the dashboard.

10. Asset Checkout Component

Purpose

Tracks assets used during events.

Responsibilities

- Allocate approved assets.
- Record asset usage.
- Track return process.

Functional Behaviour

- Approved assets are allocated before the event.
- Asset status changes to **In Use**.
- Asset usage is recorded throughout the event.
- Club Head returns assets after event completion.
- Club President verifies returned assets.
- Asset status changes back to **Available**.

11. Certificate Management Component

Purpose

Generates certificates for eligible participants and volunteers after successful completion of an event.

Responsibilities

- Verify eligibility.
- Process approval workflow.
- Generate certificates.
- Store certificates.
- Provide certificate downloads.

Functional Behaviour

- Event attendance is verified.
- Volunteer task completion is verified.
- Club Head requests certificate generation.
- Club President digitally signs certificates.
- Faculty Coordinator digitally signs certificates.
- System generates certificates.
- Users receive notifications.
- Certificates are stored in user dashboards for download.

12. Report Generation Component

Purpose

Generates analytical reports after the completion of an academic term.

Responsibilities

- Compile club analytics.
- Generate performance reports.
- Provide downloadable reports.

Functional Behaviour

- Academic term or event concludes.
- System collects club, event, volunteer, attendance, and resource data.
- Performance analytics are generated.
- Reports are prepared.
- Faculty Coordinator downloads or views reports.
- Historical reports remain available for future reference.

13. Resource Request Component

Purpose

Processes requests for introducing new venues or assets into the system.

Responsibilities

- Receive requests.
- Review requests.
- Approve or reject requests.
- Add approved resources to the system.

Functional Behaviour

- Club President submits a request for a new venue or asset.
- Faculty Coordinator reviews the request.
- Request is approved or rejected.
- Approved resources are added to the resource inventory.
- Newly added resources become available for future event planning.

Component Relationships:

Component	Direct Interaction
User Authentication	Dashboard
Dashboard	Club Management, Event Management, Volunteer Management, Task Management, Certificate Management
Club Management	Event Management
Event Management	Venue & Asset Management, Volunteer Management, QR Attendance
Venue & Asset Management	Asset Checkout
Volunteer Management	Task Management
Task Management	Proof of Work Verification, Messaging
Proof of Work Verification	Certificate Management
QR Attendance	Certificate Management
Report Generation	Club Management, Event Management, Volunteer Management, Asset Checkout, Certificate Management
Resource Request	Venue & Asset Management